

OFF-LINE SCATTERING RESONANCES FOR THE HECKE TRIANGLE ORBIFOLDS, AND WHAT THE GENERIC SCATTERING AXIOMS CANNOT DECIDE

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ABSTRACT. For every finite integer $q \geq 3$ we prove that the scalar trivial-character scattering determinant φ_q of the one-cusp Hecke triangle orbifold $G_q \backslash \mathbb{H}$ has infinitely many nonreal zeros ρ with $\Re \rho > 1/2$, and therefore infinitely many multiplicity-matched scattering poles $1 - \rho$ with $\Re(1 - \rho) < 1/2$. In particular every nonarithmetic finite Hecke group G_q , $q \notin \{3, 4, 6\}$, has a scattering resonance strictly off $\Re s = 1/2$.

We then isolate the hypotheses the proof actually consumes as an explicit axiom list A (meromorphic continuation of order at most two, the functional equation $\varphi(s)\varphi(1-s) = 1$, reality, a generalized Dirichlet series with the Hejhal archimedean factor at $\kappa = 1$, finiteness and reality of the right divisor with strip confinement, a polynomial vertical bound, and the exact critical-line modulus), and prove a metatheorem: *A entails the negation* of the naive on-line rigidity statement. No proof schema quantifying over all models of A can place all right-half-plane zeros on $\Re s = 1/2$; a schema that appears to do so is refuted already by the modular case φ_3 . We also show that A fails to decide the genuine RH-analogue $P_{\text{line}}(3/4)$ in at least one direction, unconditionally.

The moral is not new: since Davenport–Heilbronn (1936) it has been Dirichlet-series folklore that a functional equation without an Euler product does not force on-line zeros, and the Selberg class encodes that folklore as an axiom. What was missing was a theorem-strength analogue on the spectral/scattering side. This paper supplies one.

1. INTRODUCTION — THE HONEST FRAMING

1.1. What is claimed, in one paragraph. Following the defensible framing fixed for this paper: the moral — functional equation without arithmetic does not force on-line zeros — has been Dirichlet-series folklore since 1936 and is codified in the Selberg-class axioms. It had **no theorem-strength analogue on the spectral/scattering side**, which is precisely why spectral programs continued as though their setting were exempt. We supply that analogue: a family of scattering problems with the full analytic apparatus and provably infinitely many off-line resonances.

Mandatory correction. An earlier framing of this claim ended “with arithmeticity as the exact discriminant.” That clause is refuted by our own results and is deleted here: the non-arithmeticity clause is non-discriminating — $q = 3$ (arithmetic) has the same off-line property — and must never be used as an arithmeticity signature (see Corollary 3.4 below, which makes the blindness formal). The introduction states the blindness explicitly rather than letting a reader infer a dichotomy.

Where the novelty actually sits. The two-sided comparison — one theorem for the structure-rich side, one for the structure-poor side — already exists at theorem strength in the Dirichlet-series world: Hardy (1914) [16] proves infinitely many zeros of ζ *on* the line, Davenport–Heilbronn (1936) produces zeros *off* the line for a function without an Euler product. But those two theorems speak about **unrelated functions from different constructions**; the comparison is made by the reader, not by a parameter. The defensible novelty here is that this is the first **single-family** version of that comparison in the geometric/spectral setting: one parameter q , one construction, where a proven classification (Takeuchi’s list of the arithmetic G_q [17]) sits beside the present theorem, so the family itself splits. Nothing stronger may be claimed: the split is a statement

about the two literatures meeting inside one family, **not** an arithmeticity criterion — the off-line conclusion of §2 holds for arithmetic $q = 3$ as well.

1.2. Prior art, cited as prior art.

- **Davenport–Heilbronn (1936); Epstein zeta functions of class number > 1** [1]. The canonical negative control on the Dirichlet-series side: a function with almost all the properties of ζ except an Euler product, with zeros off the line. Cited as the origin of the moral, not as something we extend.
- **Beurling generalized primes** (Beurling 1937; the Diamond–Zhang lineage) [2]. The series-side sibling programme: invented number systems that keep part of the multiplicative structure and ask which PNT-facts survive, with known failures of the RH-analogue among them. Cited as classical companion context for the same question asked one construction over; we make no specific claim about any Beurling system and consume nothing from that literature.
- **The Selberg class.** The Euler product is an axiom there *precisely because* of Davenport–Heilbronn. Our axiom list A is the spectral-side mirror of that codification.
- **Conrey–Li (2000)** [3]. The methodological precedent: a negative control that kills a named proof schema (de Branges positivity). Our metatheorem is modelled on this argument shape.
- **Sarnak, “Problems of the Millennium: RH”** [4]. Three quotable positions: scepticism that generic self-adjointness is the source of a proof; the Conrey–Li refutation of de Branges’ positivity; the separation of “arithmetic π ” from “transcendental π ” for GRH.
- **Hejhal, LNM 1001, Thm. 7.11 / Cor. 7.12 (pp. 577–579)** [5]. The *printed partial antecedent*: for all sufficiently large N , zeros and poles of φ_N in any prescribed rectangle touching the critical line. Weaker in q -range, stronger in localization. Cited wherever novelty is framed.
- **Selberg (1990), “Remarks on the distribution of poles of Eisenstein series”** [6]. The source of the $d = 2$ counting theorem consumed as (C), per Kelmer Remark 0.2. **Not read** by the authors of the underlying notes; see the declarations of unread sources, §6.
- **Kelmer, arXiv:1402.4780** [7], §4 — the complex-analytic proof template (torsion-free global setup; only the template is reused).
- **Friedman–Jorgenson–Smajlović, arXiv:2011.12795** [8], §§2.1, 2.4 — continuation, functional equation, divisor list in orbifold generality.
- **Venkov, Trudy Mat. Inst. Steklov 153 (1981), Thm. 3.5, p. 59** [9] — **not** the 1979 Uspekhi survey; reached only through FJS.
- **Mayer–Mühlenbruch–Strömberg** [10], transfer operator for the Hecke triangle groups — presentation and the one-cusp statement.

1.3. What is *not* claimed.

- We do **not** claim there is no proof of RH without an Euler product.
- We do **not** claim the RH-analogue in this family is unprovable, undecidable, or independent of anything.
- We do **not** claim any arithmeticity criterion. Corollary 3.4 is a *blindness* statement.
- P_{naive} failing is a statement about a naive formulation, not about RH.

One scope sentence to forestall an apparent conflict (theta group). A reader who recalls Hejhal Theorem 7.11 may suspect tension with axiom A7, because that proof turns on $|\varphi_\infty| \neq 1$ for the theta group G_∞ . There is none: G_∞ has two cusps and therefore lies outside the scope fixed by A0 and by our standing restriction to finite $q \geq 3$ with one cusp.

2. THE LAW (MAIN THEOREM)

Theorem 2.1 (paper-level, unconditional). *For every finite integer $q \geq 3$, the scalar trivial-character scattering determinant of the one-cusp Hecke triangle orbifold has infinitely many nonreal zeros ρ with $\Re \rho > 1/2$, and therefore infinitely many multiplicity-matched scattering poles $1 - \rho$ with $\Re(1 - \rho) < 1/2$. In particular, every nonarithmetic finite Hecke group G_q , $q \notin \{3, 4, 6\}$, has a scattering resonance strictly off $\Re s = 1/2$.*

Supporting count, in the form the existence conclusion needs:

$$F_q(\tfrac{1}{2}, T) = \frac{1}{4\pi} T^2 \log T + O_q(T^2),$$

finitely many total right zeros forcing the defining sum to be only $O_q(T)$.

Certified-as-absent, and stated as absent here. No effective first height; no q -uniform error; no machine formalization of the analytic content; no project-specific Selberg-zeta normalization. These remain OPEN and are not needed for the stated LAW.

Three wording repairs. φ_q has no central divisor and the normalization L_q^* has the exactly simple zero at $s = 1/2$; the horizontal $O_q(\log T)$ term uses the right-edge normalization and the Selberg–Titchmarsh argument as reproduced by Kelmer, not the modulus bound alone; no explicit formula for the group-dependent linear coefficient A_q is consumed, only its existence.

Consume-side warning. Do **not** consume A_q , B_q , or C_q numerics from Kelmer — his printed B_Γ carries a spurious $\log \pi$ and his A_Γ assembly formula is wrong (see the erratum, §4.1).

2.1. The two certified pins at $q = 5$. Beside the LAW’s infinitude statement, two individual Selberg-zeta zeros of G_5 are interval-certified to contour standard (Arb argument principle, directed rounding, winding = 1 so each zero is simple).

Theorem 2.2 (two pins; computer-assisted). Z_{G_5} has zeros s_1, s_2 with

$$\begin{aligned} \Re s_1 &\in [0.45389418007494470, 0.45389618007494470], & \Im s_1 &\in [5.7635362417301305, 5.7635382417301305]; \\ \Re s_2 &\in [0.41054273549473627, 0.41054473549473627], & \Im s_2 &\in [7.81976724701551188, 7.81976924701551188]. \end{aligned}$$

*The closed real-part intervals are disjoint with separation ≥ 0.04334944458020843 . Consequently the nonreal strip zeros of Z_{G_5} lie on **no single vertical line** $\Re s = c$, for any $c \in \mathbb{R}$.*

Sources: the first-pin assembly (pin 1, DECLARED 2026-08-15, five adversarial rounds) and the second-pin assembly (pin 2, REFEREED — PROMOTED 2026-08-26; two cold seats, both PASS-WITH-CORRECTIONS, corrections applied). The no-single-line consequence is Corollary 9.1 below; its logical core is machine-verified (§5, `TwoPinNoLine.lean`). By the FJS completed-zeta divisor classification (one cusp, scalar φ_5) and the Lean-verified pole-to-zero reflection under $\varphi(s)\varphi(1-s) = 1$, the pins transport to two zeros $\rho_i = 1 - s_i$ of φ_5 with distinct real parts in $(1/2, 1)$ — the witness of §3.10 consumes. Dependency standing: computer-assisted + citation-backed (FJS/MMS); including the MMS $q = 5$ heading caveat and the FJS p. 4 k -notation caveat, both stated in full at their point of use below.

3. THE AXIOM LIST A AND THE METATHEOREMS

3.1. The structures. A *Hecke-type scattering pair* is $M = (\varphi, \mathcal{D})$ with φ meromorphic on $\mathbb{C}$ and $\mathcal{D} = (d(n), g_n)_{n \geq 1}$ its Dirichlet data. Write $\mathfrak{M}(A)$ for the class of pairs satisfying A0–A7. Nothing in the language of A mentions q , a group, a surface, arithmeticity, or ζ .

3.2. The axioms.

- A0** (scalar, one channel, $\kappa = 1$). φ is a 1×1 determinant; degree of singularity 1; the archimedean factor carries $\kappa = 1$.
- A1** (meromorphy and right-half-plane regularity). φ meromorphic on $\mathbb{C}$ of order at most 2, holomorphic in $\Re s > 1/2$ apart from finitely many poles.
- A2** (functional equation). $\varphi(s)\varphi(1-s) = 1$.
- A3** (reality). $\varphi(\bar{s}) = \overline{\varphi(s)}$; equivalently $d(n) \in \mathbb{R}$.
- A4** (generalized Dirichlet series, Hejhal archimedean factor). For $\Re s > 1$,

$$\varphi(s) = \sqrt{\pi} \frac{\Gamma(s - \frac{1}{2})}{\Gamma(s)} \sum_{n \geq 1} d(n) g_n^{-2s},$$

with $0 < g_1 < g_2 < \dots \rightarrow \infty$ discrete, $d(1) \neq 0$, the series absolutely convergent. **A4⁺** (used only where flagged): $d(n) > 0$.

- A5** (right divisor: finiteness, reality of poles, strip confinement — D2-corrected form). In $\Re s > 1/2$: finitely many real zeros $\rho_i > 1/2$ with multiplicity; every pole of φ in $\Re s > 1/2$ is real and lies in $(1/2, 1]$, and there are finitely many; and every zero lies in a vertical strip. (The reality-of-poles clause is used essentially.)
- A6** (vertical polynomial bound). For every $\varepsilon > 0$ there is $C(\varepsilon) < \infty$ with $|\varphi(\sigma + it)| \leq C(\varepsilon)$ for $1/2 \leq \sigma \leq 3/2$, $|t| \geq \varepsilon$.
- A7** (critical-line modulus). $|\varphi(1/2 + it)| = 1$ for real t , in the exact-modulus form, not merely unitarity.

Not in A, deliberately: any Euler product, multiplicativity, or Ramanujan bound; any group, surface, or spectral input beyond A0; any q -uniformity; any effective first height; any arithmeticity.

Note on strip confinement. Strip confinement is not an independent assumption: it is an immediate corollary of A4 via the right-edge estimate $|L^* - 1| < 1$ for large $\Re s$.

3.2.1. *Where φ_q sits.* φ_q is **not** a member of the Selberg class. For $q = 3$ this is unconditional and elementary: $\varphi_3 = \Lambda(2s-1)/\Lambda(2s)$ has a pole at every $s = \rho/2$ with ρ a nontrivial zero of ζ , so no $(s-1)^m \varphi_3(s)$ is entire, and φ_3 is unimodular on $\Re s = 1/2$, which no Selberg-class function is. For general q the frequencies g_n are not integers, so even the Dirichlet-series axiom fails as stated. To our knowledge φ_q also lies outside the known extensions of the class, but we have not attempted a systematic check and make no classification claim.

3.3. Breadth lemma. $\varphi_q \in \mathfrak{M}(A)$ for every finite integer $q \geq 3$, arithmetic and non-arithmetic alike. Receipts are per-axiom, both sides; in five of the nine rows the two columns are the same citation (A0, A4, A4⁺, A6, A7); A1, A2, A3, A5 carry distinct arithmetic-side derivations from Λ . A was not assembled by intersecting two separately verified lists.

For $q = 3$ the Dirichlet data is explicitly $d(n) = \phi_{\text{Euler}}(n)$, $g_n = n$, $d(1) = 1$, giving $\sum d(n) g_n^{-2s} = \zeta(2s-1)/\zeta(2s) = L_3^*(s)$. Hejhal's Lemma 7.3, used in the A4⁺ row, is printed for $N \geq 4$ and does **not** cover $q = 3$; $q = 3$ is covered by the direct positive-integer count.

3.4. The two candidate rigidity statements.

$$P_{\text{naive}} : \varphi \text{ has no zero } \rho \text{ with } \Re \rho > 1/2 \text{ and } \Im \rho \neq 0,$$

$$P_{\text{line}}(c) : \text{every zero } \rho \text{ with } 1/2 < \Re \rho < 1, \Im \rho \neq 0, \text{ has } \Re \rho = c.$$

This is the canonical definition of P_{naive} and must be used everywhere; A5 explicitly permits finitely many *real* right zeros, so the “no zeros at all” reading is non-derivable for a second and trivial reason.

3.5. Metatheorem I.

Metatheorem 3.1. $A \models \neg P_{\text{naive}}$. Every $M \in \mathfrak{M}(A)$ has infinitely many zeros ρ with $\Re \rho > 1/2$, $\Im \rho \neq 0$, and hence by A2 infinitely many multiplicity-matched poles $1 - \rho$ with $\Re(1 - \rho) < 1/2$.

Consequence (the no-go). There is no valid derivation of P_{naive} from A — not because A is too weak, but because A proves its negation. Any argument that appears to derive on-line rigidity in the sense of P_{naive} from meromorphic continuation, the functional equation, critical-line unitarity, a generalized Dirichlet series, and polynomial vertical growth **contains an error**, and the error can be exhibited: **apply the argument to φ_3** , where the failure is classical fact.

Proof sketch. This is the LAW read as a statement about $\mathfrak{M}(A)$ rather than about Hecke orbifolds. Every input is an axiom: the right-edge estimate from A4; the Jensen/Littlewood rectangle from A1, A5 (finitely many real right poles), A6 (horizontal edges), A4 (right edge to $+\infty$); the critical-line integral with leading coefficient $1/4\pi$ from A7’s exact modulus plus $|\Gamma(\frac{1}{2} + it)/\Gamma(it)|^2 = |t| \tanh(\pi|t|)$; the divergence step from A5 against $F(\frac{1}{2}, T) = (1/4\pi)T^2 \log T + O(T^2)$, with strip confinement (A5) consumed at the divergence step to convert an unbounded weighted sum into infinitely many zeros; strictness from the vanishing Jensen weight at $\Re \rho = 1/2$, independently from A2+A3+A7 forcing no divisor on the line at all; nonreality from A5; and reflection from A2. No step consumes a group, a surface, arithmeticity, an Euler product, or q .

Status and the exact residual. The unconditional form $A \models \neg P_{\text{naive}}$ is licensed by a genericity/transfer addendum: the Sel90-bypass derivation cites only interface facts plus classical ambient analysis, each a consequence of the corrected axiom list, so the derivation transfers verbatim to an arbitrary $M \in \mathfrak{M}(A)$. Residuals that remain, none of them a hypothesis on φ : the rectangle identity is consumed only in the (J)-avg/H3 form; GAP-1 ((J)-sharp, remainder $O_q(\log T)$ at every height) and GAP-2 ((C) and (DIF)) still rest on Selberg 1990 and are absent from the conclusion chain’s signature.

3.6. Metatheorem II.

Metatheorem 3.2 (conditional on RH). Assume RH. Then $A \not\models \neg P_{\text{line}}(3/4)$: the pair $\varphi_3 \in \mathfrak{M}(A)$ satisfies $P_{\text{line}}(3/4)$. Hence there is no valid derivation from A of the failure of on-line rigidity in its RH-analogue form.

The conditionality is unavoidable for this witness; we give no argument that no other $M \in \mathfrak{M}(A)$ has unconditionally collinear right-strip zeros.

3.7. The RH calibration.

Proposition 3.3. $P_{\text{line}}(3/4)$ holds for φ_3 if and only if RH. Moreover P_{naive} fails for φ_3 unconditionally.

Proof. From $\varphi_3(s) = \Lambda(2s - 1)/\Lambda(2s)$: the zeros of φ_3 in $\Re s > 1/2$ are exactly $s = (1 + \rho)/2$ for ρ a nontrivial zero of ζ , a multiplicity-preserving bijection, all nonreal, all with $1/2 < \Re s < 1$; and $\Re((1 + \rho)/2) = 3/4 \iff \beta = 1/2$. Confirmed unconditionally and re-verified numerically to 20+ digits. The closed form for φ_3 is a **not-read** standard citation (Iwaniec, *Spectral Methods of Automorphic Forms*, 2nd ed., §3.4 [11]), corroborated twice inside our own bank.

Why this matters. The functional equation reflects about $\Re s = 1/2$, so $1/2$ is the *symmetry* line; the zeros of the underlying ζ are pushed to $\Re s = 3/4$ by $w = 2s - 1$, so $3/4$ is the *rigidity* line. Writing “on-line rigidity” without saying which line is the mistake this paper exists to catch.

By A2, $P_{\text{line}}(3/4)$ for zeros is equivalent to all nonreal scattering poles of the modular orbifold lying on $\Re s = 1/4$ — the Faddeev–Pavlov / Lax–Phillips reading (both **not-read** citations here [12, 13]; the equivalence above does not depend on them).

3.8. The decision table.

statement	relation to A	status
P_{naive}	$A \models \neg P_{\text{naive}}$	PROVED, unconditional in the upgraded form; residuals GAP-1/GAP-2 outside the conclusion chain
$\neg P_{\text{line}}(3/4)$	$A \not\models \neg P_{\text{line}}(3/4)$	PROVED, conditional on RH; witness φ_3 ; conditionality unavoidable for this witness
$P_{\text{line}}(3/4)$	$A \models P_{\text{line}}(3/4)?$	SETTLED NO (2026-08-26) — Metatheorem III gives $A \not\models P_{\text{line}}(3/4)$ unconditionally (was OPEN/RH-hard)
decidability of $P_{\text{line}}(3/4)$ by A	A fails to decide it in at least one direction	PROVED UNCONDITIONALLY
$P_{\text{line}}(c)$, every $c \in (1/2, 1)$	$A \not\models P_{\text{line}}(c)$ for every c simultaneously	PROVED (Metatheorem III); computer-assisted/citation standing; witness M_5 ; unconditional, Sel90-free

The fifth row (2026-08-26) is the sharpest statement in the paper: $A \not\models P_{\text{line}}(3/4)$ holds **unconditionally**, so the third row’s question is settled negatively without RH.

3.9. Corollary — arithmeticity-blindness, correctly glossed.

Corollary 3.4. *If $A \models S$ then S holds for φ_q for every finite $q \geq 3$. Consequently **no consequence of A separates arithmetic $q \in \{3, 4, 6\}$ from non-arithmetic q , and no A -only argument can serve as an arithmeticity criterion.***

Mandatory accompanying sentence. The Dirichlet *data* is **not** arithmeticity-blind: the structures are pairs $(\varphi, \mathcal{D})$ with the g_n equal to the $|c|$ -values, which lie in $\mathbb{Z}[\lambda_q]$ and therefore encode q . An argument that inspects the Dirichlet data — still “generic analytic machinery” by any ordinary reading — is **not** covered by this corollary. Any gloss of the form “ A cannot separate the arithmetic members from the non-arithmetic ones” is refuted as written.

3.10. Metatheorem III — the former open problem, discharged. The problem as originally posed: exhibit $M = (\varphi, \mathcal{D}) \in \mathfrak{M}(A)$ and two nonreal zeros ρ_1, ρ_2 with $1/2 < \Re \rho_i < 1$ and $\Re \rho_1 \neq \Re \rho_2$. It is now discharged by the §2.1 pins.

Metatheorem 3.5 (fixed-witness quantifiers explicit). *There exists one $M_5 = (\varphi_5, \mathcal{D}_5) \in \mathfrak{M}(A)$ — $\mathcal{D}_5$ the Hejhal/FJS Dirichlet data of the $q = 5$ A_4 receipt — such that for every $c \in (1/2, 1)$, M_5 does not satisfy $P_{\text{line}}(c)$. Hence for every $c \in (1/2, 1)$, $A \not\models P_{\text{line}}(c)$: no member of the family $P_{\text{line}}(c)$ is derivable from A . One common countermodel serves all c .*

Standing (part of the statement). *This result is computer-assisted and citation-backed: it rests on the Arb interval certificates for the two pins, the Lean-verified logical joints of §5, and the cited published theorems — the FJS divisor classification (with the k -notation caveat of the dependency ledger) and MMS Theorem 6.4 at $q = 5$ (with the heading caveat stated there). The scope is exactly the family $P_{\text{line}}(c)$ — not every statement informally describable as “on-line rigidity.”*

How each of the three original blockers fell: the Selberg-zeta pins transport to φ_5 zeros $\rho_i = 1 - s_i$ by the FJS divisor classification + one-cusp scalar specialization + the Lean-verified reflection (§2.1) — an indirect, citation-backed certificate the cold referee ruled sufficient; the second certified pin (2026-08-26) supplies the distinct real part; and no synthetic Davenport–Heilbronn countermodel is needed. Membership $M_5 \in \mathfrak{M}(A)$ is the breadth lemma at $q = 5$. **Sel90-independence.** The pin chain does not consume the LAW, so Metatheorem III is free of the [Sel90, Lemmas 1, 2] residual carried by Metatheorem I — independence from that named engine, not from the Selberg/scattering literature at large. Since RH is literally $P_{\text{line}}(3/4)$ for φ_3 , Metatheorem III is the licensed form of the slogan: any derivation of any $P_{\text{line}}(c)$ from the shared axioms alone contains an error, exhibitable by running it on M_5 .

3.10.1. *Proof of Metatheorem III — the two-pin witness.* Let $\mathcal{D}_5 = (d_5(n), g_{5,n})_{n \geq 1}$ be the Hejhal/FJS Dirichlet data used in the $q = 5$ A4 receipt, and set $M_5 = (\varphi_5, \mathcal{D}_5)$ — the scalar scattering determinant of the Hecke triangle group G_5 with its Dirichlet-series data. $\mathfrak{M}(A)$ contains pairs, not bare functions.

Membership. $M_5 \in \mathfrak{M}(A)$ by the breadth lemma, at the caveat level below.

Two zeros, distinct real parts. From the two certified Selberg-zeta pins via the cited FJS divisor step, followed by the Lean-verified order-preserving pole-to-zero implication under $\varphi(s)\varphi(1-s) = 1$ (the verified artifact is the nested file `Scat1Lemma31Reflection.lean` — Lean formalizes *only* that implication, not FJS, MMS, or φ_5 itself):

$$\rho_1 = 1 - s_1 : \Re \in [0.54610381992505530, 0.54610581992505530], \quad \Im \in [-5.7635382417301305, -5.7635362417301305]$$

$$\rho_2 = 1 - s_2 : \Re \in [0.58945526450526373, 0.58945726450526373], \quad \Im \in [-7.81976924701551188, -7.81976724701551188]$$

No conjugation enters: $\rho_i = 1 - s_i$. Both nonreal (imaginary intervals strictly negative), both real-part intervals strictly inside $(1/2, 1)$, closed-interval separation exactly $0.04334944458020843 > 0$, hence $\Re \rho_1 \neq \Re \rho_2$ certified.

For any fixed c , at least one of $\Re \rho_1, \Re \rho_2$ differs from c , so a certified zero of φ_5 violates $P_{\text{line}}(c)$. Consequently no derivation from A alone can establish any member of the family $P_{\text{line}}(c)$ — the genuine RH-analogue family, not merely P_{naive} .

The Lean file `TwoPinNoLine.lean` additionally machine-verifies the pure logical core (`no_common_line`: two distinct-real-part zeros refute every vertical line simultaneously; exact gap $4334944458020843/10^{17}$; disjointness; distinct- $\Re$), sorry-free, axioms [`propext`, `Classical.choice`, `Quot.sound`].

Why this route is free of the Sel90 debt. Metatheorem I consumes the LAW, which carries the undischarged [Sel90, Lemmas 1, 2] citation. This exhibition does **not** consume the LAW: the pin chain is contour certificates (machine) + the Hilbert→Banach transport R5 + MMS Theorem 6.4 + whole-box K_s exclusion + FJS divisor + Lean reflection core; the membership proof is the breadth lemma’s row-by-row Hejhal/FJS/MMS receipt table. Independence means the named [Sel90, Lemmas 1, 2] engine is absent — the route still cites other published Selberg/scattering results and is **not** citation-free.

Dependency ledger (carried caveats).

- A5 flag superseded; A4 discreteness source-established by FJS Thm. 2.1 (enumeration = corroboration); A1/A6 Hejhal/FJS transcriptions cold re-extracted, not machine-checked quotations; width-one normalization changes φ by the zero-free factor c^{1-2s} (functional equation and divisor preserved; disclosed).
- **MMS $q = 5$ heading inconsistency:** eq. (34) heading prints “ $q = 2h_q + 3 > 5$ ” while the paper’s general odd- q incidence formulas, Lemma 6.3, Theorem 6.4, and the explicit $q = 5$ functional equation use odd $q \geq 5$; the $q = 5$ identification rests on the general incidence formula ($h_q = 1$ specialization independently checked against the three-row engine code). Any use of MMS Theorem 6.4 at $q = 5$ must carry this internal heading inconsistency.

- **FJS p. 4 notation inconsistency:** immediately after defining $k = \sum_j \dim V_j$ (degree of singularity, $= 1$ here), FJS prints “ $k = 0$ ” in a sentence evidently colliding with automorphic-weight notation; Theorem 2.1 and the divisor formulas retain the degree-of-singularity parameter. Does not alter the set-level divisor classification; disclosed wherever FJS is cited as the scalar-specialization source.
- Both pin assemblies are computer-assisted; the FJS bypass is an INDIRECT, citation-backed φ_5 zero certificate (no direct evaluator of φ_5 exists); the cold referee ruled that closure of this open problem requires zeros of φ , not a specified certification technology, so this meets the standard.

3.11. A worked audit — how to apply Metatheorem I to a published construction. A no-go of this kind is only useful if a reader can run it. This subsection runs it once, in full, on a named construction from the literature, so the procedure and — equally important — its limits are both visible.

Ledger discipline, binding on every sentence below. Metatheorem I licenses exactly one form of conclusion: *any derivation of P_{naive} whose every step is available in an arbitrary $M \in \mathfrak{M}(A)$ contains an error, and the error is exhibitable by running the derivation on φ_3 .* It licenses nothing about the correctness of any particular paper. Accordingly we do **not** claim that the construction audited below is wrong, that its Hamiltonian fails to exist, that its self-adjointness claim is false, or that its programme cannot succeed. The audit’s only output is a **burden statement**: name the step that is not available in a general $M \in \mathfrak{M}(A)$. A construction that can name one is untouched by this paper.

3.11.1. The test case. We use the Berry–Keating xp lineage and, as its most visible recent instance, Bender, Brody and Müller, “Hamiltonian for the zeros of the Riemann zeta function,” Phys. Rev. Lett. **118**, 130201 (2017) (arXiv:1608.03679) [14]. As reported in that paper’s abstract, the authors construct a Hamiltonian whose eigenvalues, subject to a boundary condition on the eigenfunctions, correspond to the nontrivial zeros of ζ ; the classical limit reproduces the Berry–Keating xp picture; the operator is not Hermitian in the given inner product but is PT -symmetric with broken PT symmetry, and the authors give a *heuristic* construction of a metric operator defining an inner product in which the Hamiltonian would be Hermitian. The paper’s own closing statement of the gap is explicit and is the reason it is the right test case: **a rigorous proof that this Hamiltonian is self-adjoint would establish the Riemann hypothesis.** This line is classified elsewhere in our bank as exposure class (b) with a (c) tail, its open gap being “precisely a *generic* self-adjointness claim.”

We audit the **self-adjointness step only** — the step that is open — and not the construction of the operator, which is manifestly ζ -specific. We have not verified the paper’s internal details beyond its abstract and our classification of it; every description above is flagged “as reported in [BBM17]” and the audit is written schematically so that it does not depend on those details.

3.11.2. Step (i) — what the argument uses about the target function. The self-adjointness step, stripped to the properties of the target function it consumes, uses (as reported in [BBM17] and in the surrounding xp literature):

- (1) that the target is a meromorphic function of one complex variable with a reflection symmetry about a distinguished vertical line;
- (2) that its relevant divisor is discrete, of finite density in horizontal strips, and confined to a vertical strip;
- (3) that the reflection symmetry acts on the divisor as an involution pairing ρ with its mirror image, together with the reality symmetry pairing ρ with $\bar{\rho}$;
- (4) that the boundary condition on the eigenfunctions is a scalar, one-channel condition;

- (5) that the resulting spectral problem is unitary on the distinguished line, in the sense that the associated scattering/transfer quantity has modulus one there;
- (6) polynomial vertical growth in the closed strip, used to control the asymptotic and the metric-operator series.

The conclusion sought is that the eigenvalues are real, i.e. that the divisor lies on the distinguished line.

3.11.3. *Step (ii) — each item is a consequence of A or of ambient analysis.*

item	supplied by
1. meromorphy with reflection about a distinguished line	A1 + A2
2. discreteness, finite strip density, strip confinement	A5, with strip confinement itself an A4 corollary; finite density in strips from A1 (order ≤ 2) via Jensen
3. reflection and reality involutions on the divisor	A2 + A3
4. scalar, one-channel boundary data	A0
5. unitarity on the distinguished line	A7, in the stronger exact-modulus form
6. polynomial vertical growth in the closed strip	A6
ambient toolkit (Stirling, Jensen/Littlewood, Phragmén–Lindelöf, Schwarz reflection, sub-harmonicity)	classical analysis, model-independent

No item on the list mentions primes, an Euler product, multiplicativity, a Ramanujan bound, a group, a surface, or arithmeticity. That is the substance of the audit: the properties the self-adjointness step is reported to use are all in A .

3.11.4. *Step (iii) — the conclusion Metatheorem I licenses.* Suppose the self-adjointness step could be completed using only items 1–6 and ambient analysis. Then the completed argument is a derivation, from premises each of which holds in every $M \in \mathfrak{M}(A)$, of the statement that the divisor in the right half-plane lies on the distinguished line — that is, of P_{naive} . By Metatheorem 3.1, $A \models \neg P_{\text{naive}}$. Hence such a completion is impossible, and any argument that appears to achieve it contains an error. Moreover the error is **exhibitable**, and cheaply: instantiate the argument at $M = (\varphi_3, \mathcal{D}_3) \in \mathfrak{M}(A)$, where items 1–6 all hold and the conclusion is false as classical fact; the first step of the argument that fails there is the step that was never available.

The burden this places is therefore precise and small: **name the step of the self-adjointness argument that is not available for a general $M \in \mathfrak{M}(A)$** — equivalently, the step at which ζ -specific arithmetic re-enters. If such a step can be named, the argument is untouched by this paper and has merely been disciplined. If it cannot, the argument is incomplete in a way that no amount of additional analytic rigour will repair.

3.11.5. *What this audit does not establish — four explicit weakenings.*

- (1) **No refutation of [BBM17].** We do not claim the construction is wrong. The audit’s conclusion is conditional on the self-adjointness step being completable from items 1–6 alone, which is precisely what is open. Nothing here bears on the correctness of the Hamiltonian, the PT analysis, or the heuristic metric.
- (2) **The construction *is* ζ -specific; only the open step is audited.** The operator in [BBM17] is built from ζ -data, so the programme as a whole is not an A -only schema. If the metric operator’s rigorous construction turns out to consume the Euler product, the

explicit formula over primes, or any other arithmetic input, then the audit does not bite, and we do not assert that it does.

- (3) **Items 1–6 are our reconstruction, not the paper’s own hypothesis list.** We have not read a rigorous statement of the intended self-adjointness argument, because none is published; we reconstruct the properties it would use from the abstract and from the xp literature. A published argument using a property outside A escapes the audit automatically.
- (4) **The transfer to $\mathfrak{M}(A)$ requires the schema to be restatable.** Metatheorem I quantifies over Hecke-type scattering pairs; an operator-theoretic argument bites only once it has been restated as a statement about $(\varphi, \mathcal{D})$. Where a construction resists that restatement, the audit is inapplicable rather than inconclusive. We record this as a genuine limitation of the method, not as a formality.

A fifth caveat is inherited from Corollary 3.4: an argument that inspects the **Dirichlet data** $\mathcal{D}$ — the g_n , which lie in $\mathbb{Z}[\lambda_q]$ and encode q — is still “generic analytic machinery” by any ordinary reading, and is nonetheless outside the reach of the arithmeticity-blindness corollary. The audit above must be read as covering arguments that use φ and the axioms, not arguments that mine $\mathcal{D}$.

4. APPENDIX A (SUMMARY) — THE JENSEN/LITTLEWOOD RECTANGLE, RE-DERIVED

Purpose: self-containedness. The rectangle identity (J) that the counting argument consumes is normally imported from Selberg 1990, Lemmas 1 and 2, reached only through Kelmer’s (4.20). Appendix A re-derives it from the standard complex-analytic toolkit and the interface facts already used.

form	statement	status
(J)-sharp	rectangle identity with remainder $O_q(\log T)$ at every T	PARTIALLY re-derived — one residual, GAP-1, a pointwise negative-part bound at height T
(J)-avg	same identity, remainder $O_q(\log T)$ for some $T^* \in [T - 1, T]$	RE-DERIVED, unconditional on the banked inputs
H3	$F_q(1/2, T) = (1/4\pi)T^2 \log T + O_q(T^2)$, per- q , both halves	RE-DERIVED from (J)-avg plus monotonicity of F_q

Since the LAW’s existence conclusion needs only H3, **the [Sel90] citation is replaceable for the theorem of §2.** It is *not* replaceable for the sharp asymptotic (C), which consumes (J)-sharp through the finite-difference step; (C) does not appear in this paper’s conclusion chain.

Appendix contents in outline: Lemma A (orientation calibration of the Γ -integral), Lemma B / (LW ∞) with the $1/\pi$, Lemma C (the crux — the potential $P(s) = -\int_s^\infty \log L^*$ has purely imaginary jumps across the cuts, so $\Re P$ is continuous and branch-free; no zero count is used anywhere, hence no circularity), and Lemma D (conformal sub-mean-value on a translated half-disc, with $\kappa(R)$ independent of T because $H = H_0 + iT$ is a pure translate). The referee reproduced Lemma C’s numerics on independent code at dps = 30, agreeing to 10^{-26} – 10^{-32} at seven heights including two where the divisor is empty and two straddling the first zero ordinate $\gamma = 7.06736257086735$.

Honesty items. GAP-1 blocks only (J)-sharp; the Fubini justification is by compactness on the box for each fixed T , with no uniformity in T claimed or needed; the $3/2 < \sigma < \sigma_1$ strip is covered by the absolutely convergent series bound.

4.1. Remark — an erratum in Kelmer, arXiv:1402.4780, eq. (4.18). Printed as a short remark, framed as service to the literature. Kelmer’s (4.18) evaluates its last term as $\log |t \tanh(\pi t)/\pi|$, where in fact $|\Gamma(\frac{1}{2} + it)/\Gamma(it)|^2 = t \tanh(\pi t)$ exactly. The $/\pi$ inside the logarithm injects a spurious $-\frac{1}{2} \log \pi$, so the printed constant $B_\Gamma = (-4 \log \pi - 1)/(8\pi) \approx -0.22198$ is wrong. Direct numerical evaluation of

$$\frac{1}{2\pi} \int_{-T}^T (T - |t|) \log |L^*(\tfrac{1}{2} + it)| dt - \frac{1}{4\pi} T^2 \log T,$$

divided by T^2 , gives -0.2107732 at $T = 200$, -0.2105234 at $T = 1000$ and -0.2104734 at $T = 5000$, converging to $(-2 \log \pi - 3)/(8\pi) \approx -0.2104609172$. The printed A_Γ assembly formula is likewise wrong; the corrected relation is $A = a + 2B$, the coefficient D cancelling identically.

The constant identities of that correction — the finite-difference leading term, the identity $A = a + 2B$ with D cancelling, the corrected constant, and its disequality from Kelmer’s printed B_Γ (which reduces to $\log \pi \neq 1$) — are targets of `LawSkeletonI.lean` (Aristotle dispatch v33) and are machine-verified there, sorry-free and axiom-clean.

What this remark does not do. The argument of this paper consumes **none** of Kelmer’s A_q , B_q , C_q constants (see the consume-side warning of §2), so nothing above is load-bearing for any statement here. The erratum is reported because it is in the printed literature, not because we depend on it.

5. MACHINE VERIFICATION

5.1. How the claims of this paper were checked. Documentation of process, not a claim of method. Comparable verification pipelines exist elsewhere; nothing in this subsection is offered as a methodological novelty. It is recorded only so a referee can see what was done.

- **Cold adversarial refereeing.** Every load-bearing claim passed at least one cold adversarial referee pass on an independent lineage, with re-derivation rather than transcription: the LAW twice, the metatheorem twice, and the Jensen/Littlewood re-derivation once.
- **Independent numerics.** The Appendix A crux was re-run on separately written code at $\text{dps} = 30$, agreeing with the author’s values to 10^{-26} – 10^{-32} (26–31 digits) at seven heights, four of them never tested by the author.
- **Hash-pinned primary sources.** Each consumed PDF is recorded with its digest; e.g. arXiv:1402.4780 at sha256 c15fb0c4d1d72cc1e09ee6c70532e27d835afd8a8e01a23668cdb6049f8d5030. Sources reached only through another author’s transcription are declared as such in §6.
- **Machine verification of the combinatorial finish** in Lean 4, axiom-clean and conditional on the named hypotheses, as detailed below.
- **Append-only corrections.** Every correction was applied as a dated append-only block, leaving the superseded text in place, so the full audit trail is recoverable in the project repository.

5.2. The verified statement. The combinatorial / real-analytic **finish** of the theorem of §2 is machine-verified in Lean 4 (Mathlib), conditional on named hypotheses H1–H5, in `aristotle_dispatch_v33_aris` (the returned, sorry-free artifact; status line “No sorry bodies remain,” with an independent local re-compile recorded). The verified statement is: growth of the weighted Jensen count together with finiteness of the real zeros implies infinitely many strictly off-line zeros and their reflections. **No scattering-theoretic content is machine-verified:** φ_q has no Lean definition, and no spectral theory, meromorphic continuation, Jensen/Littlewood rectangle, or property of φ_q is formalized. Those enter only as named hypotheses.

Two pointer corrections that must be respected:

- (1) The *dispatch* file `aristotle_dispatch_v33/LawSkeletonI.lean` is *not* the artifact: it carries 16 `sorry` occurrences and states “This file machine-verifies nothing.” Cite the `aristotle_dispatch_v33_aristotle/` path.
- (2) Per the dispatch summary, rows S5 (the rectangle (J)) and H3 (`hgrowth`) are relabelled to PROVED (refereed 2026-08-23, corrections D-1..D-3 applied), in the consumed (J)-avg form only. Scope limits unchanged: no q -uniform constant, no effective first height, no machine formalization of the analytic content. **H4 and H5 keep their “NOT proved here” labels.**
- (3) The logical core of Metatheorem III / Corollary 9.1 is machine-verified in `aristotle_dispatch_v34/project` (returned artifact; local re-elaboration “Build completed successfully (8034 jobs)”); four theorems — `no_common_line`, exact interval gap $4334944458020843/10^{17}$, disjointness, distinct real parts — sorry-free, axioms exactly [`propext`, `Classical.choice`, `Quot.sound`]; statements byte-identical to the dispatch). The same scope warning applies: φ_5 has no Lean definition; FJS/MMS content is cited, not formalized. The pole-to-zero reflection core is the v33 nested artifact `Scat1Lemma31Reflection.lean` (root dispatch stub still carries `sorry`; cite the nested file).

5.3. Data availability. Draft paragraph, to be finalized at submission. All material supporting this paper is archived in the project repository: the Lean 4 sources and their compilation logs; the interval-arithmetic certificates and shard-level receipts of the localization computation (§7), including the per-leaf certification records and the cross-host determinism check; the numerical scripts and outputs underlying Appendix A and the erratum of §4.1, together with the independently written referee code and its values; the dated, append-only correction blocks and the full cold-referee reports for every load-bearing claim; and the SHA-256 digests of every primary source consumed.

6. DECLARATIONS OF UNREAD AND TRANSCRIBED SOURCES

- **Selberg 1990, Lemmas 1 and 2** — the complex-analytic engine citation, reached only through Kelmer’s transcription of (4.20). **Not read** by any author or referee of the underlying notes. Its content is re-derived in Appendix A in the consumed form; GAP-1 and GAP-2 still rest on it and are outside the conclusion chain.
- **Selberg 1990** is also the correct attribution for the $d = 2$ counting theorem consumed as (C) (per Kelmer Remark 0.2); Kelmer’s own contribution is the $d \geq 3$ generalization.
- **Iwaniec, §3.4** — the classical closed form of φ_3 ; declared not read, corroborated twice inside our bank.
- **Venkov, Trudy Mat. Inst. Steklov 153 (1981), Thm. 3.5, p. 59** — reached only through FJS, never at source. The 1979 Uspekhi survey is the wrong item.
- **Hejhal §7 and FJS** — consumed through `pdftotext transcriptions`, not verbatim quotations; the A1 (order ≤ 2) and A6 chains rest on these.
- **Faddeev–Pavlov, Lax–Phillips** — not read; cited for context only.
- Hejhal (7.2)–(7.5) are stated for the conjugated group with cusp width 1 and $\varkappa \equiv 1$; the normalization changes φ only by the zero-free factor c^{1-2s} , which preserves the divisor and the functional equation. A is stated for the conjugated normalization.

7. CERTIFIED LOCALIZATION AT $q = 8$ — DEPTH-8 CHECKER OUTPUT

The merged depth-8 campaign records the following bounded computational facts:

- All 1024/1024 depth-8 leaves across 4 arcs have status PASS, satisfy $q_{Op} < 1$, and satisfy all strict gates.
- Cross-host determinism is pinned: Kaggle and local produced a byte-identical record with $rH = 0.1892125248420895230\dots$ and $qOp = 0.3271992747911403256\dots$

- The merged checkpoint was validated by the checker’s own `validate_checkpoint_records` routine.
- Checker sha: 6a9c1c3d7b28c2e0741a5e880d1b12d48066437ea03efcfd3cda90743f1fc3b0.

Mandatory scope caveat (verbatim from the merge report, does_not_mean). This is checker output, not a theorem. E1, the $q = 8$ MMS/Hilbert identification, K_s , analytic gates 5-6 and continuation condition 8 of the 12-item ledger are untouched and remain OPEN.

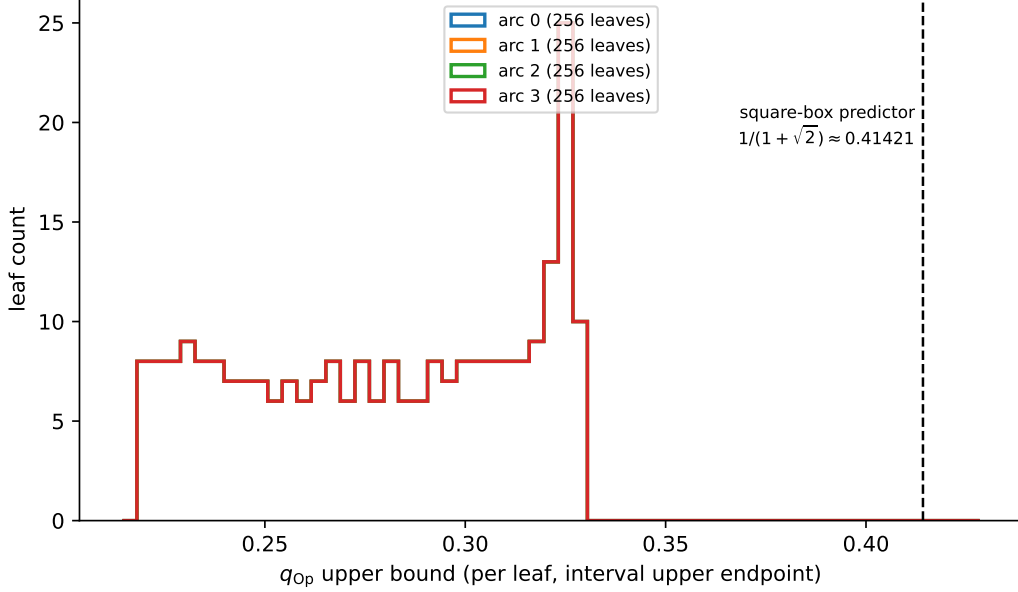


FIGURE 1. Per-arc distribution of q_{0p_upper} across the 1024 depth-8 leaves, arcs 0–3, with the square-box predictor $1/(1 + \sqrt{2}) \approx 0.41421$ marked. **Not yet rendered.**

8. REMARK / OUTLOOK — PRIME GEODESICS (NOT A THEOREM)

Remark. The off-line resonances of §2 are Selberg-zeta zeros — by the Friedman–Jorgenson–Smajlović divisor description (subject to the FJS k -notation caveat of the dependency ledger) they enter *reflected*, at $1 - \rho$ with $\text{Re} < 1/2$. Their total contribution to the prime geodesic counting formula is therefore $O_q(x^{1/2}(\log x)^2)$: dominated by the main term and by every error term under study. The correct statement is *structural bookkeeping*, not a constraint: the compact-surface divisor description of Selberg-zeta zeros fails for every finite $q \geq 3$ (non-arithmetic in particular; the property is non-discriminating — $q = 3$ has it too), by an infinite, counted margin ($\gg_q T \log T$ reflected zeros below height T) — harmlessly for the PGT. Printed partial antecedent: Hejhal LNM 1001, Thm. 7.11/Cor. 7.12 (large N).

Why the two sides are now asymmetric in kind. On the arithmetic side the divergence is already theorem-grade: improved error terms in the prime geodesic theorem for arithmetic surfaces are proven — Luo–Sarnak (1995) [15] and its successors — by exploiting the hidden multiplicative structure those surfaces carry. On the non-arithmetic side the corresponding expectations have rested on heuristics. The theorem of §2 supplies a *structural* statement on that side: infinitely many off-line scattering resonances, in the frame the explicit formula uses. That is all it supplies.

What must not be said here. We derive no prime geodesic error term, no Ω -result (the LAW gives no information on the imaginary parts, so no non-cancellation bound is reachable even

in principle), no effective first resonance height, and no q -uniform statement; the LAW gives no effective first height at all. This remark is an outlook paragraph naming a consumer, not a claim about the PGT.

9. COROLLARY — NO SINGLE VERTICAL LINE

The results of §2.1, translated into the Selberg-zeta zero language of G_5 directly, give the following.

Let G_5 be the Hecke triangle group, $\lambda_5 = 2\cos(\pi/5)$ the golden ratio; $X_5 = G_5 \backslash \mathbb{H}$ the non-arithmetic finite-area hyperbolic orbifold; $Z_S = Z_{G_5}$ its Selberg zeta function, meromorphically continued to $\mathbb{C}$. Write

$$\Lambda := \{s \in \mathbb{C} : Z_S(s) = 0\}, \quad \Lambda^\circ := \Lambda \cap \{0 < \Re(s) < 1, \Im(s) \neq 0\}$$

(the non-real strip zeros), and $L := \{s \in \mathbb{C} : \Re(s) = 1/2\}$ (the critical line). Λ° excludes the real-axis zeros (trivial zeros from the identity contribution and the small-eigenvalue parameters $s \in (1/2, 1]$), which are present for every finite-area surface and carry no information.

Corollary 9.1. Λ° contains two zeros with certified real-part intervals

$$\Re(s_1) \subset [0.45389418007494470, 0.45389618007494470], \quad \Re(s_2) \subset [0.41054273549473627, 0.41054473549473627]$$

disjoint with closed-interval separation $\geq 0.04334944458020843 > 0$. Hence **no single vertical line** V_c **contains** Λ° , **for any** $c \in \mathbb{R}$: for each fixed c at least one of the two certified real parts differs from c .

Proof. Immediate from Theorem 2.2: the certified finite winding + closed-contour exclusion, together with the finite-section identity, the Gohberg–Krein/Simon determinant comparison with machine constants [18, 19], and the argument-principle homotopy, place exactly one zero of $\det(1 - L_{s,+})$ in each box; the Hilbert→Banach transport (R5) shows both boxes lie in the domain $\Omega^* = \{\Re s > 1/2\} \cup \{\Re s > 0, \Im s > 1\}$ where the Hilbert-space and MMS Banach determinants agree; the whole-box K_s -divisor exclusion shows neither box can have its zero cancelled by the K_s factor; and MMS Theorem 6.4 (subject to the $q = 5$ heading caveat recorded in §3.10.1), $Z_S = \det(1 - L_{s,+}) \cdot \det(1 - L_{s,-}) / \det(1 - K_s)$, with $\det(1 - L_{s,-})$ analytic near each box, converts each determinant zero into a zero of Z_S of multiplicity at least that of the $+$ factor. Since $\Im(s_1), \Im(s_2) \neq 0$ and $0 < \Re(s_i) < 1$, both $s_1, s_2 \in \Lambda^\circ$. Disjointness of the real-part intervals with the stated separation is exact decimal arithmetic, independently re-verified with Arb directed rounding. For any fixed c , at least one $\Re(s_i)$ interval excludes c , so $\Lambda^\circ \not\subset V_c$. ■

Scope limits. No claim is made that Λ° is disjoint from L in general, nor any density, counting, or distribution claim, nor any family claim for G_q , $q \neq 5$ (the family statement “every non-arithmetic Hecke group has an off-line resonance” is a separate open ticket, not a consequence of this corollary), nor any “arithmetic $\iff$ critical line” law: the arithmetic Hecke triangle groups $\{3, 4, 6\}$ form one commensurability class (one data point), and the contrast noted here is descriptive of the data, not a proved equivalence. The result is computer-assisted, resting on machine-certified Arb/Acb interval receipts, Lean-proved abstract joints (Aristotle, axiom-clean), cited published theorems (Simon [18], Grothendieck’s Résumé [19], MMS Theorem 6.4 [10], standard Selberg/scattering theory), and short self-contained paper-proofs. The resonance *interpretation* is citation-level standard theory, not re-proved here; the determinant *localization* does not depend on it.

10. ACKNOWLEDGMENTS AND AUTHORSHIP

DRAFTING CHECKLIST BEFORE ANY SUBMISSION

- ☐ Owner decides authorship and whether to invite a coauthor.
- ☐ Every “NOT READ” declaration of §6 survives into the submitted text.
- ☐ Hejhal Thm. 7.11 / Cor. 7.12 cited wherever novelty is framed.

- ☐ The non-arithmeticity clause never presented as a discriminant.
- ☐ §7 either carries a completed, refereed certificate or is deleted.
- ☐ Bombieri cited at most once, and not as a no-go.
- ☐ Zenodo (or equivalent) deposit created and its DOI substituted into the data-availability paragraph.
- ☐ The worked audit re-checked against the published text of every construction it names; the four weakenings survive into the submitted text, and no sentence asserts that a named paper is wrong.
- ☐ A cold referee reads the assembled manuscript before submission.

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